TOR DIMENSION OF A COMPLEX ON AN AFFINE SCHEME EQUALS TOR DIMENSION OF THE CORRESPONDING MODULE COMPLEX

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Allowed standard prerequisites (editorial).

- On an affine scheme the derived category of modules identifies with the derived category of quasi-coherent sheaves by the tilde construction, compatible with tensor product. used throughout at proof lines 2522-2549. Lets the sheaf statement be translated into module statements and back.
- A complex of modules with tor amplitude in [a, b] is represented by a complex of flat modules concentrated in [a, b]. invoked at proof lines 2525-2528. Produces the flat representative whose sheafification witnesses the tor amplitude of E.
- A complex of flat sheaves concentrated in degrees [a, b] has tor amplitude in [a, b]. invoked at proof lines 2531-2533. Concludes the forward direction of (1) from the flat representative.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Used throughout the proof as the affine dictionary (proof lines 2522-2549); the two directions of the tor amplitude equivalence are inside the counted span.

Source reference. Original lemma 0654: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 2526-2529 to replace M^bullet by a flat representative; the sheafification step and conclusion are inside the counted span.

Source reference. Original lemma 08CI: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 2531-2533 to conclude the forward direction; the flat representative it is applied to is constructed inside the counted span.

Lemma 0.2. *Let $X = \mathrm{Spec}(A)$ be an affine scheme. Let $M^\bullet$ be a complex of A-modules and let E be the corresponding object of $D(\mathcal{O}_X)$. Then*

(1) *E has tor amplitude in $[a, b]$ if and only if $M^\bullet$ has tor amplitude in $[a, b]$.*
(2) *E has finite tor dimension if and only if $M^\bullet$ has finite tor dimension.*

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Proof. Part (2) follows trivially from part (1). In the proof of (1) we will use the equivalence $D(A) = D_{QCoh}(X)$ of Lemma 06Z0 without further mention. Assume $M^\bullet$ has tor amplitude in $[a, b]$. Then $K^\bullet$ is isomorphic in $D(A)$ to a complex $K^\bullet$ of flat A-modules with $K^i = 0$ for $i \notin [a, b]$, see More on Algebra, Lemma 0654. Then E is isomorphic to $\widetilde{K^\bullet}$. Since each $\widetilde{K^i}$ is a flat $\mathcal{O}_X$-module, we see that E has tor amplitude in $[a, b]$ by Cohomology, Lemma 08CI.

Assume that E has tor amplitude in $[a, b]$. Then E is bounded whence $M^\bullet$ is in $K^-(A)$. Thus we may replace $M^\bullet$ by a bounded above complex of A-modules. We may even choose a projective resolution and assume that $M^\bullet$ is a bounded above complex of free A-modules. Then for any A-module N we have

$$E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \widetilde{N} \cong \widetilde{M^\bullet} \otimes^{\mathbf{L}}_{\mathcal{O}_X} \widetilde{N} \cong \widetilde{M^\bullet \otimes_A N}$$

in $D(\mathcal{O}_X)$. Thus the vanishing of cohomology sheaves of the left hand side implies $M^\bullet$ has tor amplitude in $[a, b]$. □